WebClient Computing

Javascript Basics

Department of Computer Science

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Announcement

- Exam April 22nd 10:30 in Room 7호관 231
- The graded first homework is out
- If you did not show your kmu-id via slack private message, please do!
- The homework-2 is due by April 21st before the class time
  - Encourage to use tags that were covered in the slides
- The project submission due is by May 1st
  - In the project, you are encouraged to use tag/styles not covered in the class

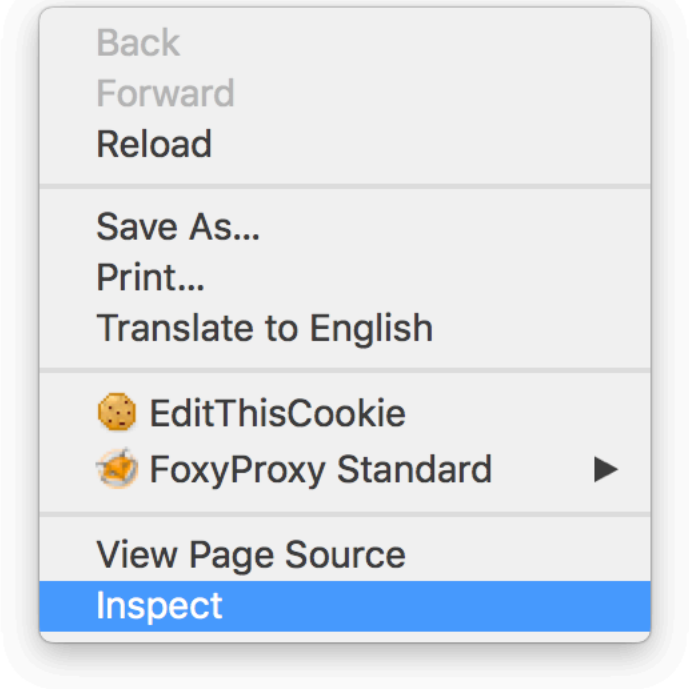
How to Load Javascript File into a Browser

- In a HTML file header

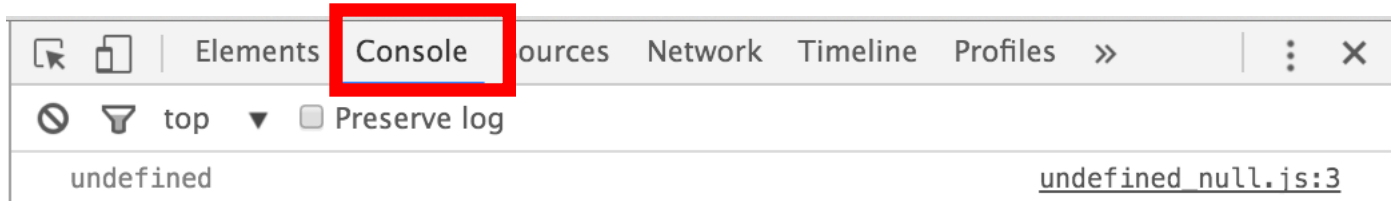
```
<head>
  <title>Javascript exercise</title>
  <script type="text/javascript" src="undefined_null.js"></script>
</head>
```

Open Javascript Console

- Right click on a Chrome browser



- In the Panel, click Console tab



Type Conversion From Strings

- parseInt
- parseFloat

```
var num = prompt("Type a number to convert: ");
console.log(num, typeof(num));
var num_int = parseInt(num);
console.log(num_int, typeof(num_int));
var num_float = parseFloat(num);
console.log(num_float, typeof(num_float));
~
```

```
123.456 string                                  parse_int.js:2
123 "number"                                    parse_int.js:4
123.456 "number"                                parse_int.js:6
> |
```

Arithmetic Operators

- Addition
  - +
- Subtract
  - -
- Multiplication
  - *
- Division
  - /

Example

- binary_op.js

```
var a = 9;
var b = 5;

console.log(a+b);
console.log(a-b);
console.log(a*b);
console.log(a/b);
```

```
14                    binary_op.js:4
4                     binary_op.js:5
45                    binary_op.js:6
1.8                   binary_op.js:7
```

Arithmetic Operators

- Modulo
  - %
  - Remainder value (9%5 = 4)
- Increment operator
  - ++
  - Location matters
    - Before variable – increment first and do other operation (++a)
    - After variable - do other operation then increment (a++)
- Decrement operator
  - --
  - Location matters

Example

- increment.js

```
var a = 9;
var b = 5;

console.log(a % b);

console.log(a);
console.log(a++);
console.log(a);

console.log(b);
console.log(++b);
console.log(b);
```

4	increment.js:4
9	increment.js:6
9	increment.js:7
10	increment.js:8
5	increment.js:10
6	increment.js:11
6	increment.js:12

Javascript Object

- Javascript object is similar to class in Java
  - Person class
    - Name
    - Age
    - Hometown
    - …
- Object definition is wrapped with {}
- Object attribute and value is separated by colon :
- Multiple attribute-value pair is separated by comma ,
- EX: var instructor = {name:“kyungyong”, age:21};
- Object Property – a member of an object with a value
- Object method – a function to perform execution with given inputs

Javascript Object Example

- object.js

```
var teacher = {name:"kyungyong", age:21};
console.log(teacher.name);
console.log(teacher["age"]);

teacher["name"] = "leeky";
teacher.age = 22;
console.log(teacher);
~
```

```
kyungyong                                   object.js:2
21                                          object.js:3
Object {name: "leeky", age: 22}             object.js:7
```

Javascript Functions

- Object has
  - Property – a variable with a value
  - Method - function
- Given a set of input arguments perform a set of operations and return output
- Helps abstraction (추상화)
  - Users gives inputs to a function and do not care how the inputs are handled
- How it is shaped
  - *function* name(arg1, arg2) {
  - ….
  - }
- The function body can reference the input arguments
- return keyword
  - Stop the operation and exits the function with or without a value

Functions Example

- functions.js

```javascript
function test_return() {
  console.log("in the test_return function");
  return;
  console.log("return is already called");
}

function test_print(message) {
  console.log("in the test_print function");
  console.log(message);
  console.log("exiting the test_print function");
  return;
}

function sum(arg1, arg2) {
  var output = arg1 + arg2;
  return output;
}

test_return();
test_print("Learning Javascript");
var a = sum(3,4);
console.log(a);
```

```
in the test_return function                    function.js:2
in the test_print function                     function.js:8
Learning Javascript                            function.js:9
exiting the test_print function                function.js:10
7                                              function.js:22
```

String Concatenation Member Access

- Merges multiple strings into one
  - string.concat(another_string)
  - string1 + string2
- String Object members
  - string.length – get the length of a string
  - string.concat(string2) – merge two strings
  - string.charAt(position) – get a string at a specific position
  - string.substring(startPosition, endPosition) – returns a sub-string from startPosition index (inclusive) till the endPosition (exclusive)
    - “Hello”.substring(1,4) ➔ “ell”
- String method references
  - http://www.w3schools.com/jsref/jsref_obj_string.asp

String Example

- string_object.js

```
var str1 = "Hello";
var str2 = " World";
var str3 = "!!";

console.log(str1.length);
console.log(str1["length"]);

var str4 = str1.concat(str2);
console.log(str4);

console.log(str1.concat(str2).concat(str3));
console.log(str1 + str2 + str3);

console.log(str1.charAt(1));
console.log(str1.substring(1, 4));
```

```
5                                   string_object.js:5
5                                   string_object.js:6
Hello World                         string_object.js:9
Hello World!!                       string_object.js:11
Hello World!!                       string_object.js:12
e                                   string_object.js:14
ell                                 string_object.js:15
```

Some Handy Math Functions

- Math.abs(x)
  - Get absolute value
- Math.power(x,y)
  - x^y
- Math.random()
  - Generate a random number between 0.0 ~ 1.0
- Math.sqrt(x)
  - Get squared-root of x
- Full reference
  - http://www.w3schools.com/jsref/jsref_obj_math.asp

Example

- math.js

```
console.log(Math.abs(-10));
console.log(Math.pow(2,3));
console.log(Math.sqrt(36));
console.log(Math.random());
console.log(Math.random());
```

```
10                                  math.js:1
8                                   math.js:2
6                                   math.js:3
0.9545098192596186                  math.js:4
0.5898707910795844                  math.js:5
```

Operators in Javascript

- Relational operator
  - An operator to express relation between two variables
  - Return value is true or false
  - > : greater than
  - < : less than
  - >= : greater than or equal to
  - <= : less than or equal to
  - == : same
  - != : different

Relational Operator Example

- rel_op.js

```
var a = 123;
var b = 456;
var c = 123;

var a_gt_b = a > b;
console.log(a_gt_b);
console.log(a < b);
console.log(a < c);
console.log(a <= c);
console.log(a == b);
console.log(a == c);
console.log(a != b);
console.log(a != c);
```

```
false                    rel_op.js:6
true                     rel_op.js:7
false                    rel_op.js:8
true                     rel_op.js:9
false                    rel_op.js:10
true                     rel_op.js:11
true                     rel_op.js:12
false                    rel_op.js:13
```

Reference

- Javascript, the definitive guide
- W3Schools Javascript